DAY 1

Seeing the forest from space

Satellite basics, Earth Engine, and how to make an AI write your code.

2 hours · Foundations + Land Cover

Labs 1 and 2 · No coding experience needed

BEFORE WE START

The next two hours

1. **Pre-test** — 10 minutes, does not count for anything. It measures what the course teaches, not what you know.
2. **How satellites see forests** — pixels, bands, and the one trick that makes vegetation mapping possible.
3. **What Earth Engine is** — and why you will never download a satellite image.
4. **Vibe coding** — how to get an AI to write correct Earth Engine code, and how to tell when it hasn't.
5. **Lab 1 and Lab 2** — your first map, and your first real number: hectares of forest in a province you choose.

You will not learn to write code this week.

You will learn to **specify** what you want precisely enough that a machine writes it correctly — and to **check** the answer well enough that you would sign your name to it. The second half is the hard half, and it is the half that stays valuable.

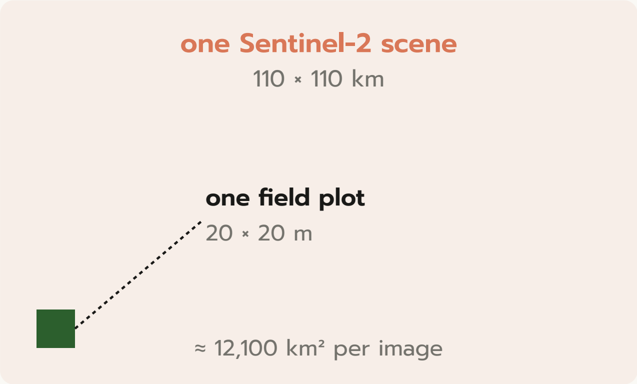
PART ONE

How a satellite sees a forest

Four ideas, a motivation, and a family of indices you will meet again this week.

THE MOTIVATION

A field plot tells you about 400 m². A forest is a million hectares.



one field plot
400 m² per plot

one Sentinel-2 scene
≈ 12,100 km² per image

≈ 30 million plots would fit in a single image

WHY THE PLOT ALONE IS NOT ENOUGH

A field team measuring one hectare properly — diameter, height, species — can take days. Steep terrain, flooded wetland and dangerous ground put whole regions out of reach, and a typical ground survey samples well under 1% of a landscape.

WHAT A SATELLITE BUYS YOU

Revisit — Sentinel-2 returns every 5 days. **Archive** — Landsat has recorded the same places since 1972. **Coverage** — one command answers for an entire province, not a sample of it.

IDEA 1 OF 4

An image is a grid of measurements

A satellite image is not a photograph. It is a **grid of numbers**, and each number is a physical measurement of how much light bounced off one patch of ground.

One patch = one **pixel**. For Sentinel-2, one pixel covers **10 × 10 metres** — about the size of a badminton court.

Which means a single pixel cannot tell you about one tree. It tells you about the twenty trees standing in that square.

RESOLUTION YOU WILL MEET THIS WEEK

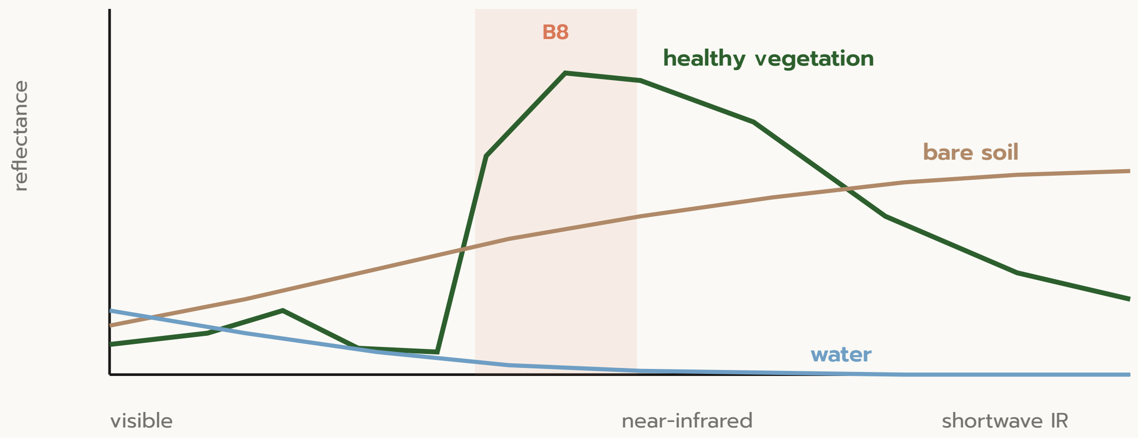
Sentinel-2	10 m	forest patches
Landsat	30 m	history to 1984
MODIS	500 m	fire, whole regions
CHIRPS	5.5 km	rainfall

Smaller pixels are not automatically better. They cost more computing and often answer the question no better.

Each image has many layers, one per colour

Your eye sees three colours. A satellite measures **thirteen**, including several your eye cannot see at all. Each one is a **band**.

The band that matters most for forestry is **B8, near-infrared**. Living leaves reflect it extremely strongly. Nothing else in a landscape does.



WHY THIS MATTERS

Healthy vegetation is the brightest thing in the scene at NIR, and fairly dark in red. That gap is large, measurable, and unique — which is what makes it possible to map forests from orbit at all.

An index turns bands into one useful number

NDVI — the Normalised Difference Vegetation Index — is the whole of vegetation remote sensing in one line:

$$\text{NDVI} = \frac{(\text{NIR} - \text{Red})}{(\text{NIR} + \text{Red})}$$

Bright in NIR and dark in red gives a big positive number. That is a plant. The subtraction measures the gap; the division cancels out how sunny it was.

READING AN NDVI VALUE

-1 to 0	water, cloud shadow
0 to 0.2	bare soil, rock, buildings
0.2 to 0.5	grass, crops, sparse cover
0.7 to 0.9	**healthy forest**

NDVI can never exceed 1. If yours does, something is wrong — and today you will see exactly what.

Clouds are the real enemy, and the median beats them

Thailand is cloudy. Any single image in the wet season is mostly white.

The fix is not to hunt for one perfect clear day. It is to take **every** image over three months, throw away the cloudy pixels, and for each pixel take the **median** of whatever is left.

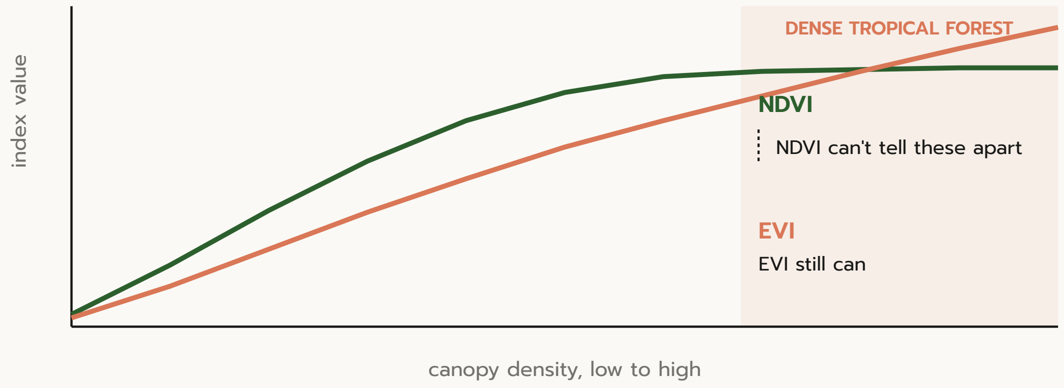
Cloud is bright and occasional; ground is darker and constant. The median lands on the ground almost every time.

THE PATTERN YOU WILL USE ALL WEEK

```
collection
  .filterBounds(aoi)         # where
  .filterDate(start, end)  # when
  .map(mask_clouds)        # drop bad pixels
  .median()                # one clean image
```

Four lines. This is the backbone of almost every satellite analysis you will ever run.

NDVI has a blind spot: it maxes out in dense forest



NDVI reads **0.85–0.95 almost everywhere** in a closed tropical canopy and runs out of headroom to tell forests apart. Adding the **blue band** — EVI — corrects for haze and buys that headroom back.

A REAL EVI MAP, VIEWED IN EO BROWSER

```
EVI = G × (NIR − Red) / (NIR + C1·Red − C2·Blue + L)
```

This is what you will actually compute in Colab.

THE NDVI FAMILY

Same recipe, different bands

NDVI is one member of a small family of indices that all follow the same shape — **(band A - band B) / (band A + band B)** — just pointed at different questions.

INDEX	BANDS	ANSWERS
NDVI	NIR, Red	how green
NDWI	NIR, SWIR	how much water is in the canopy
NBR	NIR, SWIR	how badly it burned

NDWI catches drought stress weeks before a canopy visibly browns — water content in the leaves drops long before colour does.

YOU WILL COMPUTE THIS EXACT INDEX ON DAY 3

NBR is the number behind **dNBR**, the burn-severity map you will build in Lab 5 for a real fire season in Chiang Mai.

Same formula shape as NDVI. Different bands, different question — that pattern repeats across almost everything in this field.

WHERE THIS IS GOING

From counting pixels to counting trees automatically

- **Deep learning.** Neural networks are starting to detect and count individual trees or animals in imagery directly, instead of a human tracing outlines by hand.
- **Cloud-native processing.** Petabyte-scale data and the computing to analyse it now sit in the same place — nobody downloads the imagery first.
- **Hyperspectral sensors.** Hundreds of narrow bands instead of a handful means measuring leaf chemistry — nitrogen, water content — not just greenness.

The cloud-native piece is not a future trend for you — it is **what you are about to spend the rest of today learning.** Earth Engine is exactly this: the data and the computing already sitting together, waiting for a recipe.

PART TWO

What Earth Engine actually is

The idea that makes all of this possible on a laptop.

THE MODEL

A catalog plus computers, both in the cloud

The catalog	Google's computers	You
90+ petabytes 1,000+ datasets 50 years of imagery	the processing happens there, not on your laptop	send a recipe, receive a map or a number

You never download the satellite images.

THE OLD WAY

Find a scene. Download 800 MB. Wait. Install software. Repeat for every date and tile. Run out of disk. Give up.

EARTH ENGINE

Write down what you want. Google runs it next to the data and sends back only the answer. A twenty-year analysis takes seconds.

THE CATCH

Computing is free, but it is not unlimited

Your account gets **150 EECU-hours per month**. That is generous — but it is a real budget, and it is possible to spend a lot of it in one careless cell.

Two habits keep you safe for the entire course:

- **One province, not one country**.
- **Always pass `scale=`**. Use 100 m for area statistics. Asking for 10 m gives you a hundred times the work and the same answer.

WHAT "SCALE" COSTS

SCALE	PIXELS IN A PROVINCE
`scale=10`	~120 million
`scale=30`	~14 million
`scale=100`	~1.2 million
`scale=500`	~50 thousand

For "how many hectares of forest?", all four give you the same answer to within a fraction of a percent. Only one of them is worth paying for.

PART THREE

Vibe coding

How to make an AI write correct Earth Engine code — and how to catch it when it doesn't.

The AI is fast, confident, and does not know whether it is right.

You are slower. You are also the only one in the loop who can tell. That is the job, and it does not require you to write Python.

YOUR TOOL

Antigravity — free, and split in two on purpose

Antigravity is Google's free AI assistant for this course. Install it once (Day 0), sign in with your personal Gmail, and open its chat panel.

It can also open and run files entirely on its own. **You will not use that**. You will use only the chat — ask for code, get code back, exactly like talking to Claude or ChatGPT.

1. Paste your prompt into Antigravity's chat
2. Copy the code it gives you
3. Paste it into a Colab cell and run it there

THE ONE RULE

Antigravity **writes** the code. **Colab runs it**. Copy every answer across yourself — never let Antigravity execute anything against your Earth Engine account directly.

The PROMPT recipe

P	**Purpose**	What map or number do you want?
R	**Resource**	Which dataset, by exact ID?
O	**Outline**	Which area, which dates?
M	**Must-haves**	Scale, units, palette, legend
P	**Platform**	Colab, ee + geemap, authenticated
T	**Test**	Print one number I can check

Fill all six slots and the code usually works first time.

THE SLOT EVERYONE SKIPS

Test. Without it you get a beautiful map and no way to know whether it is right. Always ask for one printed number you can check against something you already know.

THE METHOD

A real prompt, filled in

```
You are a Google Earth Engine Python expert.

PURPOSE   Build one cloud-free Sentinel-2 image of my study area.
RESOURCE  Use dataset COPERNICUS/S2_SR_HARMONIZED.
OUTLINE   Area is in the variable `aoi`. Dates 2024-11-01 to 2025-02-28.
MUST      Mask cloud and shadow with the SCL band (drop values 3, 8, 9, 10).
          Divide by 10000 for reflectance. Take the median.
PLATFORM  Google Colab, `ee` and `geemap` imported and authenticated.
TEST      Print how many scenes went into the composite.
```

Notice what is **not** in there: any Python. You described the problem in English, in the vocabulary of your own field. That is the entire skill.

WHEN IT BREAKS

The error loop

1. **STOP**. Do not change three things and re-run hoping.
2. **COPY the whole error**. Not just the last line — the lines above say which cell and which line.
3. **PASTE it back** with one sentence about what you were trying to do.
4. **ASK for corrected code**, not an explanation.
5. **Re-run and re-check**. The error going away is not proof the answer is right.

```
This is the error I got. Please give me
the corrected code.

I was trying to: calculate hectares of
forest in my province

[paste the ENTIRE red error message]
```

NEVER RETYPE CODE BY HAND

Every retyped character is a chance at a new typo. Now you are debugging two problems.

THE SKILL THAT MATTERS

The sniff test

Code that runs is not code that is correct. Before you believe any number, check it against something you already know.

QUANTITY	PLAUSIBLE	SOMETHING IS WRONG IF…
NDVI	-1 to 1; forest 0.7–0.9	**3000** — forgot to divide by 10000. **All zeros** — the mask ate everything.
Area of a class	less than your province	**More forest than land** — you summed m^2 and called them hectares.
Biomass	100–400 Mg/ha	**5** — averaging over water. **20,000** — wrong units.
Temperature	15–35 °C	**around 300** — that is Kelvin.
Accuracy	0.75–0.92	**0.99+** — your test data leaked into training.

The one question to ask every single time: is this biomass or carbon? Hectares or square metres? Celsius or Kelvin? Almost every wrong answer in this field is a units error wearing a convincing map as a disguise.

THE OTHER RISK

AI assistants invent things that look real

They are most likely to do it when the real thing is obscure — which describes most Earth Engine dataset IDs.

Three defences, in order of speed:

- Check the ID against your **cheat sheet** before running anything.
- `print(img.bandNames().getInfo())` instead of trusting a band name.
- Search the official catalog. No Google page, no dataset.

REAL EXAMPLES FROM BUILDING THIS COURSE

- `...global_forest_change_2024_v1_12`
  Real, but superseded. Current is `2025_v1_13`.
- `ee.Image('NASA/ORNL/biomass_carbon_density/v1')`
  Right ID, wrong type — it is an ImageCollection. **Every assistant we tested got this wrong**.
- A band called `NDVI` on raw Sentinel-2
  Does not exist. You compute it.

PART FOUR

Lab 1 and Lab 2

Open Colab. This is the part you came for.

LAB 1

Your first satellite map

● LIVE — follow along

1. Pick a province. Yours, not mine.
2. Ask for every Sentinel-2 image over it this dry season.
3. Throw away the cloud, take the median.
4. Turn it into NDVI.
5. **Check the number.**

WHAT YOU SHOULD GET — NAN PROVINCE

0.754
mean NDVI
heavily forested

194
Sentinel-2 scenes
in the composite

Your province will differ — that is the point. But if your NDVI is above 1, or below 0.3 in a forested area, stop and find out why before moving on.

LAB 2

From a picture to a number

A map is not a result. **A number is a result.**

Someone has already classified the whole planet's land cover. Your job is to use their map and get hectares out of it — because "how much forest is there?" is the question your job will actually ask.

```
ee.Image.pixelArea()   # m² per pixel
  .divide(1e4)         # -> hectares
  .addBands(landcover)
  .reduceRegion(grouped sum)
```

NAN PROVINCE · ESA WORLDCOVER 2021

CLASS	HECTARES	SHARE
Tree cover	1,033,493	84.4%
Grassland	126,798	10.4%
Cropland	52,639	4.3%
Built-up	7,497	0.6%
Water	3,100	0.3%

Your total must come close to the province area you printed earlier. If it is ten times off, you divided by the wrong number.

Two respected products. Same province. 52,000 hectares apart.

WorldCover says Nan is **84%** tree cover. Dynamic World says **88%**. Neither is lying.

THE LESSON

Why the two maps disagree — and what to do about it

Three honest reasons

- **Different definitions of "tree"**. Canopy height? Percent cover? Each team drew the line somewhere different.
- **Different dates**. WorldCover is a fixed 2021 snapshot. Dynamic World composites whatever dates *you* asked for.
- **Different class schemes**. WorldCover has one grassland class where Dynamic World splits grass from shrub.

SO NEVER WRITE THIS

"The forest area of Nan is 1,033,493 ha."

WRITE THIS

*"Forest covers 1,033,493 ha of Nan province **according to ESA WorldCover v200 for the year 2021**."*

The product and the year are **part of the answer**, not a footnote. This single habit will save you more embarrassment than anything else in this course.

BEFORE YOU GO

Day 1 wrap

What you can now do

- Make a cloud-free satellite image of anywhere in Thailand
- Turn it into a greenness map
- Get hectares per land cover class out of a global product
- Write a prompt that produces working code
- **Tell when the answer is wrong**

HOMEWORK — 15 MINUTES

Run Lab 1 and Lab 2 again with `PROVINCE` set to your **home province**.

Write down two numbers and bring them tomorrow:

- mean NDVI
- hectares of tree cover

Tomorrow: you stop using other people's maps and make your own — separating evergreen from deciduous forest — then put a carbon figure on it.